

Python-Based LDA MCQs

1. Which Python library provides a built-in implementation of Linear Discriminant Analysis?
 - A) NumPy
 - B) Pandas
 - C) Scikit-learn
 - D) Matplotlib
2. Which class is commonly used to implement LDA in Scikit-learn?
 - A) PCA
 - B) LinearDiscriminantAnalysis
 - C) LogisticRegression
 - D) DecisionTreeClassifier
3. What is the purpose of the fit() method in Scikit-learn's LDA?
 - A) Visualize data
 - B) Learn discriminant directions from labeled data
 - C) Normalize features
 - D) Predict labels
4. Which method is used to reduce dimensionality after fitting an LDA model?
 - A) transform()
 - B) predict()
 - C) score()
 - D) classify()
5. In Python, which library is most suitable for computing eigenvalues and eigenvectors manually?
 - A) OpenCV
 - B) NumPy
 - C) Flask
 - D) Seaborn
6. What will happen if the target labels y are not provided while fitting an LDA model?
 - A) LDA will automatically cluster data
 - B) LDA cannot be trained properly
 - C) LDA converts to PCA

- D) No effect
- 7. Which command is used to import LDA from Scikit-learn?
 - A) from sklearn.decomposition import LDA
 - B) from sklearn.lda import LDA
 - C) from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
 - D) from sklearn.linear_model import LDA
- 8. What is the output shape after applying LDA with n_components=2 to a dataset containing 100 samples?
 - A) (2,100)
 - B) (100,2)
 - C) (100,100)
 - D) (2,2)
- 9. Which NumPy function is commonly used to compute eigenvalues and eigenvectors?
 - A) np.mean()
 - B) np.linalg.eig()
 - C) np.dot()
 - D) np.var()
- 10. In an LDA workflow, why is feature scaling often performed before applying LDA?
 - A) To reduce memory usage
 - B) To ensure features contribute comparably
 - C) To increase sample size
 - D) To remove labels
- 11. Which parameter controls the number of discriminant components in Scikit-learn LDA?
 - A) n_clusters
 - B) n_neighbors
 - C) n_components
 - D) max_depth
- 12. Which matrix operation is required when implementing LDA from scratch?
 - A) Matrix inversion
 - B) Image convolution
 - C) FFT
 - D) Hashing
- 13. While implementing LDA manually, which scatter matrix represents class compactness?

- A) Between-Class Scatter Matrix
- B) Within-Class Scatter Matrix
- C) Covariance Matrix
- D) Correlation Matrix

14. Which step comes first when implementing LDA from scratch?

- A) Compute eigenvectors
- B) Compute class means
- C) Train classifier
- D) Plot decision boundary

15. After computing eigenvectors in manual LDA, what should be done next?

- A) Randomly shuffle them
- B) Sort them according to eigenvalues
- C) Delete negative values
- D) Normalize labels

16. Which NumPy function can be used to project data onto selected discriminant vectors?

- A) `np.dot()`
- B) `np.max()`
- C) `np.sum()`
- D) `np.argmax()`

17. What is the primary purpose of `lda.predict(X_test)`?

- A) Reduce dimensions
- B) Classify unseen samples
- C) Scale data
- D) Compute covariance

18. Which metric is commonly used to evaluate an LDA classifier?

- A) Accuracy
- B) RMSE
- C) MAPE
- D) MAE

19. In Scikit-learn, LDA can be used for:

- A) Classification only
- B) Dimensionality reduction only

- C) Both classification and dimensionality reduction
- D) Clustering only

21. You are asked to implement Linear Discriminant Analysis from scratch using only NumPy (without using `sklearn.discriminant_analysis.LinearDiscriminantAnalysis`). Which sequence of steps is correct?

- A) Standardize Data → Compute Class Means → Compute Within-Class Scatter Matrix → Compute Between-Class Scatter Matrix → Solve Eigenvalue Problem of $S_W^{-1}S_B$ → Select Top Eigenvectors → Project Data
- B) Compute FFT → Compute Covariance Matrix → Apply K-Means → Project Data
- C) Compute PCA Components → Apply Logistic Regression → Predict Labels
- D) Normalize Labels → Compute Decision Tree → Extract Features → Project Data

22. What is the purpose of the following line in a manual LDA implementation?

`eigvals, eigvecs = np.linalg.eig(np.linalg.inv(Sw).dot(Sb))`

- A) Compute principal components
- B) Compute discriminant directions
- C) Compute cluster centroids
- D) Compute feature scaling factors

23. Consider:

`X_lda = X.dot(W)`

What does W represent?

- A) Weight initialization matrix
- B) Matrix of selected discriminant vectors
- C) Covariance matrix
- D) Distance matrix

24. In a scratch implementation, why are eigenvectors sorted in descending order of eigenvalues?

- A) To reduce execution time
- B) To select directions with maximum discriminative power
- C) To normalize data
- D) To balance classes

25. When implementing LDA manually, what is the most likely issue if S_w is singular?

- A) Labels disappear
- B) Matrix inversion fails
- C) Data becomes nonlinear
- D) Eigenvalues become complex automatically